

EXPERIMENT 17. Split DataFrame by School Code and Find Mean, Minimum and Maximum Age

Aim:

To write a Pandas program to split a DataFrame based on **school code** and calculate the **mean, minimum, and maximum age** for each school.

Program:

```
exp17.py > ...
1  import pandas as pd
2  data = {
3      'school_code': ['S01', 'S01', 'S02', 'S02', 'S03', 'S03'],
4      'student': ['A', 'B', 'C', 'D', 'E', 'F'],
5      'age': [15, 16, 14, 15, 16, 17]
6  }
7  df = pd.DataFrame(data)
8  result = df.groupby('school_code')['age'].agg(['mean', 'min', 'max'])
9  print("Age statistics for each school:")
10 print(result)
11
```

Output

```
PS C:\Users\dhars\Documents\Query processing\lab_questions> & C:\L
on.exe "c:/Users/dhars/Documents/Query processing/lab_questions/ex
Age statistics for each school:
              mean  min  max
school_code
S01           15.5   15   16
S02           14.5   14   15
S03           16.5   16   17
```

Result

Thus, the DataFrame was successfully split according to **school code**, and the **mean, minimum, and maximum age** were calculated for each school.

EXPERIMENT 18. Split DataFrame into Groups Based on School Code and Class

Aim:

To write a Pandas program to split a DataFrame into groups based on **school code and class**.

Program:

```
CO4 > exp18.py > ...
1  import pandas as pd
2  data = {
3      'school_code': ['S01', 'S01', 'S01', 'S02', 'S02', 'S03'],
4      'class': ['10A', '10A', '10B', '10A', '10B', '10A'],
5      'student': ['A', 'B', 'C', 'D', 'E', 'F'],
6      'age': [15, 16, 15, 14, 15, 16]
7  }
8  df = pd.DataFrame(data)
9  groups = df.groupby(['school_code', 'class'])
10 for name, group in groups:
11     print("\nGroup:", name)
12     print(group)
--
```

Output

```
Group: ('S01', '10A')
  school_code class student  age
0         S01   10A      A   15
1         S01   10A      B   16

Group: ('S01', '10B')
  school_code class student  age
2         S01   10B      C   15

Group: ('S02', '10A')
  school_code class student  age
3         S02   10A      D   14

Group: ('S02', '10B')
  school_code class student  age
4         S02   10B      E   15

Group: ('S03', '10A')
  school_code class student  age
5         S03   10A      F   16
```

Result

Thus, the DataFrame was successfully divided into groups based on **school code and class** using the Pandas `groupby()` function.